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SUPERSEDING
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MILITARY SPECIFICATION

TRANSPORTER: WELDED HULL,

FOR

MOBILE FLOATING ASSAULT BRIDGE/FERRY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the amphibious transporter for the Mobile Floating Assault Bridge/Ferry (MAB). The transporter is diesel-powered, propeller-driven on water, two- or four-wheel-driven on land with two- and four-wheel steering, and has a welded aluminum-alloy hull.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

NN-P-530	- Plywood, Flat Panel.
QQ-S-781	- Strapping, Steel, Flat and Seals.
UU-T-81	- Tags, Shipping and Stock.

FSC 5420

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VV-F-800	- Fuel Oil, Diesel.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.

Military

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-W-5044	- Walkway Compound, Nonslip, and Walkway Matting, Nonslip.
MIL-S-7124	- Sealing Compound, Elastomeric, Accelerator Required, Aircraft Structure.
MIL-H-8790	- Hose Assemblies, Rubber, Hydraulic, High Pressure (3000 PSI).
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-W-45205	- Welding, Gas Metal-Arc and Gas Tungsten-Arc, Aluminum Alloys, Readily Weldable for Structures, Excluding Armor.
MIL-W-45210	- Welding, Resistance, Spot: Weldable Aluminum Alloys.
MIL-P-46046	- Preservative Fluid, Automotive Brake System and Components.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-S-52356	- Superstructure, Interior Bay, and Superstructure, End Bay, for Mobile Floating Assault Bridge/Ferry Unit.
MIL-M-52662	- Marine Drive Assembly and Marine Drive Installation Assembly for Bridge/Ferry Superstructure Transporter.
MIL-H-52702	- Hull Assembly; Welded, for Bridge/Ferry Transporter.

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STANDARDS

Federal

FED. STD. No. 595	- Colors.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-281	- Automobile, Trucks, Truck-Tractors, Trailers and Trailer Dollies, Preservation and Packaging of.
MIL-STD-448	- Test Methods for Construction and Industrial Machinery.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-1261	- Welding Procedures for Constructional Steels.
MS33540	- Safety Wiring and Cotter Pinning, General Practice for.

DRAWINGS

ME

TA13207E3400	- Transporter: Welded Hull for Mobile Floating Assault Bridge/ Ferry.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Welding Qualifications, Section IX.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. Seventh Street, Miami, FL 33125.)

NATIONAL BUREAU OF STANDARDS (NBS)

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Description. The transporter, which includes the hull and marine drive installation assembly (see 3.10 and 3.11), shall be in accordance with Top Assembly TA13207E3400, and as specified herein. The transporter provides overland transportation, and in-water flotation and propulsion, for either the MAB end bay or interior bay bridge superstructure. After entering the water, the superstructures are connected together to form a floating bridge or self-powered ferry. The transporter is a diesel-engine-powered, two- or four-wheel drive, rubber-tired vehicle; with retractable wheels and a stowable, stern-mounted propeller drive for water operation. The transporter hydraulic system accomplishes the functions of superstructure elevation, rotation, girder pinning, curb placement and end bay ramp folding and unfolding. The hydraulic system also provides power for capstan operation, wheel elevation, two- or four-wheel land steering, marine steering, elevation and position of the marine drive propeller, and power for control of the wheel well air system. The wheel well air system enables the four wheel wells to be independently air pressurized for increased load capacity, reduced operating draft in the water, and trimming the transporter during superstructure assembly/disassembly operations.

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fit, and functional interchangeability and reliability equivalency of such an item(s) for MAB field usage. At the option of the contracting officer, validated test data, sufficient to verify the performance and reliability of the proposed item(s), shall also be submitted. The Government will grant interim acceptance or reject the proposal within 15 days after receipt of all required data. The interim-accepted item(s) shall then be provided and evaluated at the supplier's expense on the first-produced transporter(s) during the first-produced transporter inspection specified herein. Final acceptance by the contracting officer shall be contingent upon test data verifying acceptable performance during the first-produced transporter inspection.

3.3 Initial production. When specified (see 6.2), the supplier shall furnish one or more transporters, within the time frame specified, for inspection as specified in 4.4.

3.4 Materials. Materials shall be as specified herein and on the applicable drawings. The surfaces of materials used shall be free of foreign matter, scale, rust, or corrosion. Unless otherwise specified (see 6.2), the age of rubber products furnished shall not exceed 24 months from date of cure to date of acceptance of the transporter by the Government. All threads shall conform to NBS Handbook H28 unless otherwise specified on the drawings. Components using metric threads shall not be furnished. Aluminum materials shall be provided with the material manufacturer's name or trademark and the applicable alloy and temper designations in accordance with the respective ASTM specifications cited on the drawings.

3.5 Environmental requirements.

3.5.1 Operating conditions. The transporter shall withstand the usage encountered during military floating assault bridge and ferry operations in inland water, and operation over primary and secondary roads, rough terrain, mud, snow, and sand, when tested as specified herein.

3.5.2 Operating temperature. The transporter engine shall start without standby preheating within 5 minutes. The transporter shall operate as specified herein, within 15 minutes after engine start, in any ambient temperature from minus 25° F to plus 125° F.

3.5.3 Storage temperature. The transporter and its components shall not be damaged when stored at air temperatures of minus 50° F to plus 160° F.

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3.1.1 Drawings. The drawings forming a part of this specification are end product drawings which delineate the requirements of the completed parts, assemblies, and end item. All dimensions shown on the drawings are based on a temperature of 68° F through 70° F. No deviation from the requirements shown on the drawing is permissible without prior approval of the contracting officer. The supplier is responsible for determining the requirement for and preparing his manufacturing drawings which, when used with his manufacturing processes, will produce the specified end item. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.1.2 Gaged dimensions. Unless otherwise specified (see 6.2), all dimensions identified on the drawings by the symbol Ⓘ shall be inspected using the Government-loaned inspection equipment (see 3.40) to determine acceptability in accordance with Quality Assurance Pamphlet P715-106. The gages are calibrated at 68° F to 70° F.

3.2 First article(s).

3.2.1 First-produced transporter(s). When specified (see 6.2), the supplier shall furnish one or more transporters as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production, that his production methods, equipment and personnel will produce transporters that comply with the requirements of this specification. Examination and tests shall be those specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.2.2 Transporter transportation data. When specified (see 6.2), a first-produced transporter shall be weighed, measured, and the center of gravity located to provide the data required for the transportation data plate.

3.2.3 Qualification of components. No deviation from the vendor items specified on the drawings is permissible without prior approval of the contracting officer. Where the supplier proposes to use source and specification control items other than those identified on the drawings, he shall furnish at his expense all pertinent data not later than 90 days after award of the contract to fully demonstrate the form,

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3.6 Reliability and maintainability. The transporter, excluding the superstructure and the marine drive, shall (1) have a minimum **Mean-Time-Between-Failure** (MTBF) of 57 hours, and (2) operate 500 hours without need of major overhaul, when operated with a superstructure conforming to MIL-S-52356 and a marine drive conforming to MIL-M-52662 and in accordance with the established operational, maintenance, and safety procedures employed in MAB land, ferry, and bridge operations as specified in 4.6.2.34.

3.7 Durability performance. The transporter shall perform all tests specified herein without failure or permanent deformation attributable to nonconforming materials, dimensions, tolerances or incorrect workmanship.

3.8 Electromagnetic interference. The electromagnetic interference emission and susceptibility characteristics of the transporter shall conform to MIL-STD-461, Class IIIA. The supplier shall be responsible for any additions or modifications to interference-producing components in order to comply with the electromagnetic provisions specified, and that such changes do not result in malfunction or reduced reliability of the components.

3.9 Fuel, lubricants and coolant. The transporter shall be serviced, tested, and furnished with standard military service products of the types and grades specified on Drawing 13207E6710. The transporter shall operate without malfunction when serviced with these products and tested as specified herein. The lubricants of the engine, transmission, wheel well blower, axle carriers, wheel planetaries, marine drive, capstans, and steering assemblies shall not show visual evidence of metallic particles when examined as specified herein.

3.10 Hull assembly. The hull assembly shall be in accordance with MIL-H-52702, and when assembled into the transporter, shall perform as specified herein. The hull assembly serves as the hull, body, chassis, and cab for the transporter. It forms a chassis for installation of all transporter machinery and provides a base for the superstructure during all phases of the transporter operations. The hull is primarily of welded aluminum-alloy construction.

3.11 Marine drive installation assembly. The marine drive installation assembly shall be in accordance with MIL-M-52662. The marine drive installation assembly mounts to the hull assembly and is driven by the transporter machinery to provide propulsion and steering for all water

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operations. When assembled onto the transporter, the marine drive installation assembly shall perform as specified herein.

3.12 Transporter assembly. The transporter shall be assembled utilizing the hull assembly and marine drive installation assembly specified in 3.10 and 3.11. Unless otherwise specified (see 6.2), the hull and marine drive installation assemblies shall be furnished by the supplier.

3.12.1 Transporter and systems assembly, installation and adjustments. The components of the various systems of the transporter shall be assembled, installed, and adjusted to assure acceptable performance of the completed transporter as specified herein. All like components shall be interchangeable with respect to installation, performance, and quality. All adjustments not requiring an operational transporter or an installed superstructure shall be completed prior to initial start-up. Care shall be exercised to assure that final adjustments which must be completed after the transporter is operational will not result in a safety hazard when the engine is started.

3.12.2 Initial start-up. Prior to starting the engine, the hydraulic system shall meet the cleanliness requirements specified in 3.32.2. The entire transporter shall be serviced in accordance with Drawing 13207E6710. The engine shall be prepared, started, and operated at all times in accordance with the engine manufacturer's run-in and operating procedures. Upon starting the engine, all final adjustments not requiring an installed superstructure shall be made, and the transporter shall be examined as specified in 4.5.1 to assure correct assembly, adjustment, and operation of all parts and components.

3.12.3 Installation of superstructure. Upon successful completion of the initial start-up as specified in 3.12.2, a superstructure conforming to MIL-S-52356 (see 3.40) shall be installed on the transporter in accordance with TM5-5420-210-12 (see 3.40) and all incompleted adjustments made. All transporter functions and interfaces affecting the superstructure shall be examined as specified in 4.5.1 to assure correct assembly, adjustment, and operation.

3.12.4 Run-in. After completion of the transporter assembly as specified herein and prior to any other tests, the transporter shall be subjected to the run-in specified in 4.6.2.3.

3.13 Land performance. When equipped with an end bay superstructure conforming to MIL-S-52356, the transporter shall meet the following performance requirements under all conditions specified herein.

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3.13.1 Speed on level roads. The transporter shall operate at a sustained speed of 40 miles per hour (mph) on a smooth, level, hard-surfaced roadway.

3.13.2 Acceleration. The transporter shall accelerate from 0 to 35 mph in 35 seconds on a smooth, level, hard-surfaced roadway when tested as specified in 4.6.2.5.

3.13.3 Drawbar pull. The drawbar pull of the transporter shall be a minimum of 30,000 pounds, 26,000 pounds, and 3200 pounds at stall, 0.5 and 15 mph, respectively, when tested as specified in 4.6.2.6.

3.14 Water performance. A two-unit ferry, constructed of two transporters and two interior bay superstructures, or one interior and one end bay superstructure, shall operate, without load, as specified herein.

3.14.1 Ferry steering. The ferry shall be fully controllable from the cab and rear control station of each transporter throughout the full engine speed range when tested as specified in 4.6.2.7.

3.14.2 Ferry speed. The unladen ferry shall operate in still water at a minimum speed of 12 feet per second (8.18 mph) with the propeller extended 8 inches, wheels retracted and wheel wells not pressurized when tested as specified in 4.6.2.8.

3.15 Transporter watertightness. The transporter shall not leak when immersed in water to a depth of 50 inches and tested as specified in 4.6.2.9.

3.16 Cab watertightness. The installed cab assembly shall not leak when tested as specified in 4.6.2.10.

3.17 Engine system. The engine assembly, controls, and instruments shall be assembled, installed, and adjusted to assure specified performance. The engine shall operate in accordance with the engine manufacturer's recommendation, and without leakage of fuel, coolant, or lubricants. The belt tensions of all belt-driven components shall be adjusted as specified on the drawings. The front and rear throttle and the emergency air shutoff controls shall be securely mounted, adjusted, operate without binding, and be free of twists and kinks. The throttle controls shall be adjusted to provide a full-open rack position at the point of maximum throttle travel, and the rear control station throttle drag shall be

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adjusted to hold in all operating positions. The fuel shut-off control shall be adjusted to assure correct operation of the fuel shut-off lever. The engine speed shall be adjusted to 600 rpm at idle and 2300 rpm at full-load governed speed in accordance with the engine manufacturer's procedure. The engine shall have a stall speed of 1800-2000 rpm when assembled to the transmission and checked in accordance with the transmission manufacturer's recommended procedures.

3.18 Fuel system. The fuel system shall store, vent, filter, and supply fuel to the cab heater, and to and from the engine, without leakage. The system shall be clean and free of all foreign matter. The shutoff valves shall provide complete fuel shutoff and the manually operated primer shall provide for positive priming of the engine fuel system. The actual volume of fuel over the reserve of 8 gallons in each tank shall be indicated on the fuel gage within 3 gallons.

3.19 Cooling system. The cooling system shall operate without leakage of coolant under all operating conditions specified herein and the coolant temperature shall not exceed 234° F. The cooling system lines shall be installed to clear adjacent components and be free of twists, kinks, or collapsed sections. Sealing around the perimeter of the engine radiator shall be installed with specific caution to prevent recirculation of engine-room air around the radiator. The cooling fan shroud and belts shall be kept adjusted as specified on the drawings.

3.20 Engine air intake system. The air-cleaner assembly shall be securely installed and the air induction system shall have no leaks or loose connections. The hoses shall be installed clear of other components and shall be free of twists, kinks, or collapsed sections.

3.21 Exhaust system. The manifolds, muffler, and exhaust pipes shall show no evidence of exhaust leakage during operations. The exhaust system shall be mounted securely and covered by insulative jacketing as specified on the drawings. The exhaust connections and hanger brackets shall be alined and fastened securely. The exhaust system shall be painted as specified on the drawings. The exhaust pipe outlet and flapper valve shall accommodate the exhaust extension assembly and permit engagement to its full cut-out depth.

3.22 Transmission system. The transmission assembly shall be adjusted in accordance with the transmission manufacturer's recommendations after installation into the transporter. The transmission controls and instruments shall be assembled, installed, and adjusted to assure specified

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performance. The transmission range selector shall be indexed to correspond with the respective control valve detent lever positions at the transmission and assure that the starter will operate in the neutral position only. The transmission throttle valve control shall be adjusted to up-shift in accordance with the transmission manufacturer's recommendations. The hydraulic retarder control shall be adjusted to assure that the retarder valve operates within its full range and returns to the "off" position. The transmission assembly shall operate without overheating or evidence of leakage of fluids or lubricants. All transmission controls shall be free of twists and kinks, operate without binding, and be securely assembled. The transmission shall provide an engine stall speed of 1800-2000 rpm and shall absorb maximum stall torque without damage when checked in accordance with the transmission manufacturer's recommended procedures. The transmission oil temperature shall not exceed 300° F during any test specified herein.

3.23 Front axle engagement. The front axle engagement control shall be adjusted to index the control position with the detent positions of the transfer case output disconnect lever. The control assembly shall be securely assembled and mounted, operate without binding, and be free of twists and kinks. The force required to engage or disengage the front axle shall be between 35 and 55 pounds at the transmission output disconnect lever.

3.24 Drive shafts and axle carriers. The drive shafts and axle carriers shall be assembled and installed in accordance with the drawing requirements. The components shall be securely assembled, correctly alined, and exhibit no abnormal noises or vibration, or leakage of lubricants throughout the entire speed range of the transporter. Lubricant temperatures shall not exceed 200° F above ambient when operated as specified in 4.6.2.18.

3.25 Wheel and suspension assemblies. The suspension assemblies shall provide for driving and steering the wheels, and damping road shocks. The suspension units shall also provide for wheel and tire retraction and retention within the wheel wells, and for wheel and tire extension and mechanical locking in the extended position, as indicated by the activation of the wheel position indicator lights. The suspension units, control arm assemblies, wheel drive shafts, inner and outer universal joints, boots, steering knuckles and planetary assemblies shall operate freely without binding, overheating, leakage of lubricants or hydraulic fluid. The installed boots shall

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not leak water. The wheels shall be alined for camber, caster, and toe-in geometry in accordance with the drawing requirements. The suspension unit shock absorbers shall be provided with lubricating oil and pressurized with nitrogen as specified on the drawings, resulting in a shock absorber piston extension of 2 to 3 inches when tested as specified in 4.6.2.19. The suspension assemblies shall be capable of raising and lowering the transporter loaded with an end bay superstructure.

3.26 Land steering system. The steering system for both front and rear wheels shall operate smoothly over the full range of steering angles without hydraulic chatter, binding, damage, or leakage of lubricant or hydraulic fluid. The tie-rod bushing seal glands shall be adjusted to prevent water leakage into the hull. The rear wheel steering lock assembly shall engage and disengage without evidence of binding, drifting, or leakage. The warning light assembly shall indicate a lack of lockpin engagement in accordance with drawing requirements. The power steering system shall enable front and rear steering of the loaded transporter from stop to stop at all engine speeds between 1000 and maximum rpm when moving on a dry, level, concrete surface in the 1-2 transmission gear ranges. The force required to steer the vehicle shall be between 11 and 15 pounds at the rim of the front steering wheel when operating between 1 and 3 mph. The steering system shall provide positive steering control for the transporter. The respective steering position indicator light shall be "on" when the front or rear wheels are steered to the left or right, and "off" when the wheels are in a straight-ahead position.

3.27 Service brake system. The brake system shall be assembled and adjusted to operate without binding, or leakage of brake fluid or air. The brake system shall be free of all foreign matter and the hydraulic portion shall be bled of all air. Pedal resistance shall not be spongy and its travel shall not exceed 2.5 inches on a power-assisted application. Pedal linkage shall be adjusted to provide from 0.5 to 1 inch of free travel.

3.27.1 Service brakes (static). The brake system shall stop and hold the transporter (equipped with an end bay superstructure) in place when headed up and down a 60 percent grade. When tested as specified in 4.6.2.21.1, there shall be no brake pedal creepage.

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3.27.2 Service brakes (dynamic and fade). The service brakes shall decelerate and stop the transporter (equipped with an end bay superstructure) from a speed of 20 mph within 30 feet from point of initial brake application, when tested as specified herein and within 40 feet from point of initial brake application when tested in accordance with 4.6.2.21.3. The service brakes shall operate without evidence of fade, grab, chatter, or fishtail in excess of 2 feet.

3.28 Parking brake system. With the front wheel drive engaged, the parking brake shall hold the transporter (with end bay superstructure) on a 40 percent grade when headed either up or down grade. The hand lever pull to set the brake shall not exceed 75 pounds. All elements of the brake control shall be securely assembled and mounted, correctly adjusted and alined, operate without binding, and be free of twists and kinks.

3.29 Marine drive system.

3.29.1 Propeller drive. When activated by either the cab or rear station control switch, the marine drive power take-off and drive shaft assemblies shall transmit torque to the marine drive assembly propeller at all engine speeds without evidence of lack of engagement control, excessive clutch slippage or overheating, unusual noise or vibration, or leakage of lubricants or water. The selector switch in the cab shall enable control of the marine drive PTO from the cab or rear station control switch in accordance with the control panel markings. The installed marine drive assembly shall provide a minimum sustained static thrust of 5000 pounds when tested as specified in 4.6.2.23.2.

3.29.2 Marine steering. The steering assembly shall provide for 360 degree steering of the marine drive lower unit in either direction when actuated by either the cab or rear station steering wheel. The steering action shall be smooth and free of binding or chatter at all elevation positions throughout the full engine speed range. There shall be no evidence of leakage of water, lubricants, or hydraulic fluid. The steering direction shall be correctly indicated on the cab and rear indicators within 5 degrees. With the engine at idle speed, the marine drive lower unit shall steer 180 degrees in either direction to a maximum of 14 turns of the cab steering wheel and a maximum of 11 turns of the rear station steering wheel.

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3.29.3 Elevation. The elevating assembly shall provide for extension and retraction of the marine drive lower unit within the full operating range established by the mechanical stops when actuated by either the cab or rear station control lever. The assembly shall operate without evidence of binding, lack of response, erratic operation, leakage of water, lubricants, or hydraulic fluid. The actual lower unit extension shall be indicated on the cab and rear elevation indicators within 1 inch.

3.29.4 Stowing. The marine drive installation assembly shall be installed to position the marine drive column in a vertical "drive" position in accordance with drawing requirements. The installation assembly shall operate smoothly, without binding or evidence of leakage, within the stow cylinder travel limits when the cab or rear station control is actuated. The stow cylinder shall mechanically lock the marine drive assembly in the "stow" and "drive" positions. The "stow" and "drive" position switches shall be adjusted to energize the respective indicator lights in the cab when the marine drive assembly is in the "stow" and "drive" positions.

3.30 Capstans. Each capstan shall provide a uniform, sustained line pull, without stalling, of 2500 pounds at engine idle and 4000 pounds at maximum engine rpm when tested as specified in 4.6.2.24. Capstan operation shall be smooth and without evidence of chatter or leakage of hydraulic fluid or lubricant.

3.31 Electrical system.

3.31.1 Battery charging system. When tested as specified in 4.6.2.25.1, the battery charging system shall provide a charging rate of 100 amperes and a regulated voltage between 25.3 to 27.3 volts at an engine speed of 1500 rpm, when the battery voltage drops to 22.0 volts or less. The ammeter shall correctly indicate operation of the battery charging system. Battery terminal connections and cable splices shall be waterproof as specified on the drawings.

3.31.2 Electrical circuitry and connections. The electrical circuitry and connections shall be assembled and installed to assure conductance of the full electrical capacity of the components connected. The full electrical insulation qualities of all dielectric components shall not be reduced by assembly and installation. Electrical connection points shall be bare and clear of paint or other dielectric materials. Electrical circuitry and connections shall be waterproof and protected from

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the effects of moisture and fungus growth where specified on the drawings. Electrical cables shall be installed without excess slack and secured to prevent slippage through the cable clamps. The electrical circuits shall operate their respective electrical functions when tested as specified in 4.6.2.25.2.

3.31.3 Headlight alinement. The headlights shall be installed and alined as specified on the drawings.

3.31.4 Windshield wiper system. The windshield wiper system shall be assembled, installed, and adjusted to provide positive control with variable smooth speed operation without binding or physical interference. The wiper blade travel shall be in accordance with the drawings.

3.31.5 Bilge pump system. Each bilge pump shall operate when the control switch is actuated and each pump, as installed, shall discharge a minimum of 30 gpm from its outlet port when tested as specified in 4.6.2.25.5.

3.31.6 Intercommunications system. The intercommunications system shall be assembled and installed to provide electrical conductivity, insulation, shielding, and waterproofness to assure correct operation of the Government-furnished intercommunication components.

3.31.7 Radio communication system. The radio communication installation assembly shall be assembled and installed to provide electrical conductivity, insulation, shielding and waterproofness to assure correct operation of the Government-furnished and -loaned radio equipment when tested as specified in 4.6.2.25.7.

3.32 Hydraulic system.

3.32.1 Hydraulic system performance. The hydraulic system shall be assembled, installed, cleaned, serviced with the specified fluid, bled of air, and adjusted to assure acceptable performance of each hydraulic function. The hydraulic pump shall provide the hydraulic power, in accordance with the drawings, to operate the transporter and superstructure hydraulic functions. The hydraulic circuitry shall transmit hydraulic power, in accordance with the drawings, to all hydraulic functions and withstand the relief valve pressure setting of each circuit without evidence of permanent deformation or external leakage. The controls and components of the hydraulic circuitry shall start, correctly operate, and stop all transporter and superstructure hydraulic functions in a

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smooth, controlled fashion without external leakage, binding, chatter, erratic response, vibration, or abnormal noise. Operating time for hydraulic functions shall be in accordance with Table I. The hydraulic pump shall be capable of developing a minimum of 1500 psi at 600 engine rpm and a minimum of 3000 psi at an engine speed of 1000 rpm when the hydraulic oil temperature is between 145° - 155° F. All hydraulic system component materials shall be compatible with the specified fluid and lubricants.

Table I. Operating Time for Hydraulic Functions

No.	Function	Elapsed Time, sec max (or min/max) Engine Speed, rpm			
		600-650	1000-1050	1500-1550	max
1.	Front wheel steering.	-	5	-	-
2.	Rear wheel steering.	-	13	-	-
3.	Rear wheel steering lock-pin, to lock.	3	-	-	-
4.	Rear wheel steering lock-pin, to unlock.	3	-	-	-
5.	Wheels, up	-	-	12/30	12/30
6.	Wheels, down.	-	42	38	38
7.	Marine drive, stow to drive.	9	-	8	7
8.	Marine drive, drive to stow.	15	-	11	9
9.	Marine drive, engage.	3	-	-	-
10.	Marine drive, disengage.	3	-	3	3
11.	Marine drive, 180 degree steering, cab/RR station.	Cab 18	-	16	16
		RR sta 10	-	6	6
12.	Marine drive, up to down.	17	-	-	-
13.	Marine drive, down to up.	20	-	-	-
14.	Capstan, front, min rpm.	10	-	14	17
15.	Capstan, rear, min rpm.	10	-	14	17
16.	Wheel well air, LF, off to in.	5	-	-	-
17.	Wheel well air, RF, off to in.	5	-	-	-
18.	Wheel well air, LR, off to in.	5	-	-	-
19.	Wheel well air, RR, off to in.	5	-	-	-

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Table I. Operating Time for Hydraulic Functions (Cont'd)

No.	Function	Elapsed Time, sec max (or min/max) Engine Speed, rpm 600-650	1000-1050	1500-1550	max
20.	Wheel well air dump valve, left, open to close.	5	-	-	-
21.	Wheel well air dump valve, right, open to close.	5	-	-	-
22.	Superstructure, up.	20	-	9	-
23.	Superstructure, down.	2/7	-	2/7	-
24.	Superstructure, stow to bridge.	25	-	25	-
25.	Superstructure, bridge to stow.	28	-	28	-
26.	Superstructure curb, up				
	(int)	14	-	14	-
	(end)	18	-	16	-
27.	Superstructure curb, down				
	(int)	22	-	20	-
	(end)	18		16	-
28.	Superstructure pins, extend.	6	-	-	-
29.	Superstructure pins, retract.	6	-	-	-
30.	Superstructure end bay, ramp extend (Avg of 10 trials).	-	70/95	40/65	-
31.	Superstructure end bay, ramp retract (Avg of 10 trials).	-	70/95	40/75	-

Notes:
1. Hydraulic fluid temperature range plus 125° F to plus 190° F.
2. Above values are valid when vehicle is on dry, hard surface only.
3. Above values are for the operation of only one function at any given time.

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3.32.2 Hydraulic system cleanliness. Prior to their installation into the transporter, the interior and exterior surfaces of all hydraulic components shall be washed and flushed throughout to remove all visible contaminants, particles, and foreign elements. Openings of the components shall be immediately covered to prevent the entry of contaminants and shall remain covered until installation. Prior to initial start-up, the assembled hydraulic system, less pump, reservoir, and marine drive hydraulic assembly shall be flushed throughout utilizing an auxiliary pump, reservoir, and 10 micron (absolute) filtration arrangement. (Note: Provision of this arrangement shall be the responsibility of the supplier.) After installation of the specified hydraulic fluid and achieving fluid contamination level within the limits specified in Table II, the remaining components shall be installed. The system shall then be operated and bled of all air prior to being subjected to the tests specified herein. The hydraulic system shall be kept clean of foreign matter and contamination during all tests. Upon completion of all tests, and prior to final acceptance, the hydraulic filter elements shall be replaced and hydraulic fluid shall meet the contamination limits specified in Table II when tested as specified in 4.6.2.

Table II. Hydraulic Fluid Contamination Limits
(Particles per milliliter)

Size Range (micron)	Particle Quantity
> 10	1600
> 20	114
> 30	17
> 40	3.8

After successfully completing the cleanliness test, the system shall be tagged at the cab hydraulic console, with a tag conforming to UU-T-81, Type A, as follows: "The hydraulic system of this transporter has passed the hydraulic cleanliness test of MIL-T-52703." The tag shall be signed by the supplier's test and quality control representatives and a Government representative.

3.32.3 Hydraulic system slaving. When provided with hydraulic slave hoses, the transporter shall be capable of slaving the hydraulic system functions of another transporter or be capable of being hydraulically slaved by another transporter without evidence of damage or leakage when tested as specified in 4.6.2.26.3.

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3.33 Pneumatic systems.

3.33.1 Compressed air system. The system shall be assembled, installed and adjusted to assure acceptable performance of the brakes, horns, tire inflator, and low air pressure warning signal without evidence of leakage, binding, or foreign matter. The system shall maintain air pressure without activating the low air pressure warning signal during simultaneous operation of the brakes and horns at engine idle. The system shall be capable of inflating the transporter tires and indicating the tire pressure when tested as specified in 4.6.2.27.1. The low air pressure switch relief valve and pressure reducing valve shall operate as specified on the drawings.

3.33.2 Wheel well air blower system. The system shall be assembled, installed, and adjusted to supply and control air to the four wheel wells, individually or in any combination. With the transporter immersed in water as specified in 4.6.2.9, the system shall completely displace the water from all four wheel wells within 3 minutes when the engine is operated at 1500 rpm. Operation of the system shall not show evidence of leakage of air or lubricating fluid. The unloading valves shall provide for air discharge at the rear axle carrier when not directed into the wheel wells. The air hoses and ducts shall be securely installed, clear of other components, free of twists, kinks, or collapsed sections. The blower belts shall be alined and tensioned in accordance with the drawing requirements. The wheel wells shall retain a minimum of 1.3 psig air pressure after 5 minutes, when tested as specified in 4.6.2.27.2.

3.34 Fire extinguisher system. The system shall be assembled, installed, and adjusted to provide acceptable performance without binding or leakage when tested as specified in 4.6.2.28. Control assembly actuation shall discharge the extinguisher cylinders and shut off the engine fuel supply. The delay valve assembly shall prevent discharge of the extinguishing agent until the engine has stopped. The cylinders shall be charged to their rated capacity and tagged at final acceptance to indicate the charge, inspection date, and inspector's and supplier's names.

3.35 Cab heating system. The heating system shall be assembled, installed, and adjusted to provide heat for the cab compartment and for defrosting the windshields. With the heater switch in the "LO" position, the heater shall ignite within 5 minutes. With the heater operating in the "HI" position, air shut off shall actuate the overheat switch and cut off the fuel flow within 15-30 seconds. With the heater operating on "HI", placing the switch into the "OFF" position shall stop the fuel

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flow immediately and shall stop the blower operation within 1.0-3.5 minutes. The heater fan shall be operable without operating the heater. There shall be no evidence of leakage of fuel or exhaust when operating the heater.

3.36 Treatment and painting. Cleaning, treatment, and painting of components shall be in accordance with the drawings and as specified herein. Painted surfaces, when tested as specified in 4.6.2.30, shall show no evidence of bare metal. Surfaces inaccessible for painting after installation, such as carbon steel hydraulic tubing, shall be painted before installation. All exterior surfaces of the assembled transporter, interior surfaces of the cab assembly, wheel wells, wheel well covers, and hull access doors and grills shall be cleaned, treated and painted in accordance with MIL-T-704, Type B, lusterless. The interior of the hull, to a height between **12 and 18** inches, and all ferrous components totally inside the transporter, except source and specification control and cadmium-plated components, shall be painted in accordance with MIL-T-704, Type B, semigloss white between Color No. 27875 and 27778 of FED. STD. No. 595. Masking to provide a well-defined paint departure line on the hull interior is not required. Specification or source control components which are installed within the transporter shall be treated and painted in accordance with the requirements specified on the respective drawings. All ferrous components, plated or unplated (except cadmium-plated fasteners totally within the transporter), shall be isolated from the aluminum hull components with sealing compound conforming to MIL-S-7124. Nonslip coating conforming to MIL-W-5044, Type II, olive drab, shall be applied as specified on the drawings.

3.37 Markings. Transporter identification markings, tire pressure, jacking points, vehicle weight classification, and all other markings shall be as specified on the drawings.

3.38 Items to be furnished with transporter. No transporter shall be shipped unless the on vehicle material specified on Drawings 13207E2970 and 13207E2974 is provided with the transporter, unless approval for shipment without such items has been received from the contracting officer.

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3.39 Government-furnished property. When specified (see 6.2), Items 1 and 2 of the following property in the quantities indicated will be furnished by the Government. Item 3 will be furnished by the Government (see 6.4).

Item No.	Description	Identification	Qty for each transporter
1.	Hull Assembly, Welded for Bridge/Ferry Transporter.	MIL-H-52702	1
2.	Marine Drive Assembly for Bridge/Ferry Superstructure Transporter.	MIL-M-52662, Type II, Class 2	1
3.	On Vehicle Material - Government-furnished and manufacturer installed.	Dwg. 13207E2974	1 set

3.40 Government-loaned property. When specified (see 6.2), any or all of the following property in the quantities indicated will be loaned by the Government (see 6.5):

Item No.	Qty	Description	Identification
1.	1	Transporter: Mobile Floating Assault Bridge/Ferry with TM5-5420-210-12.	NSN 5420-00-491-6330
2.	1	Superstructure, End Bay: Mobile Floating Assault Bridge/Ferry with TM5-5420-205-14.	NSN 5420-00-491-6320
3.	1	Superstructure, Interior Bay: Mobile Floating Assault Bridge/ Ferry with TM5-5420-205-14.	NSN 5420-00-491-6339
4.*	1	Wheel alinement equipment.	Dwg. 13207E6110
5.*	1	Adapter, gage, for wheel alinement.	Dwg. 13207E6121
6.	1	QA Pamphlet, "Care, Operation and Maintenance of Inspection Equipment for Bridge/Ferry Unit, Interior-Bay and End-Bay Amphibious: Mobile Floating Assault Bridge/Ferry Unit."	P715-106, Included in Top Assembly microfilm.

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7.	1 (Partial)	Inspection equipment for Bridge/ Ferry Unit, Welded Hull Transporter.	Quality Assurance Pamphlet, P715-106, included in Top Assembly microfilm.
8.	2 sets	On-vehicle material - Government-furnished and troop installed.	Dwg. 13207E3275
9.	1 set	Maintenance Set, Mobile Floating Assault Bridge/Ferry: Section Level.	NSN 5420-00-913-1948
10.	1 set	Maintenance Set, Mobile Floating Assault Bridge/Ferry: Company Level.	NSN 5420-00-913-1949

* In chest No. 6, Dwg. 13207E6137.

3.41 Workmanship.

3.41.1 Transporter assembly. The transporter shall be assembled in full accordance with the requirements specified on the drawings and specifications. All parts, components and assemblies containing moving parts shall be assembled free of harmful extraneous materials such as metallic particles, dirt, grit, or corrosion.

3.41.2 Damaged components. Repair of materials, parts, components, or assemblies which have sustained damage during manufacturing, transportation, storage, or installation shall not be utilized in the transporter assembly unless their repair and use is specifically approved by the contracting officer.

3.41.3 Castings and forgings. Castings and forgings which are parts, components, and assemblies of the transporter shall be in accordance with the specification specified on the drawings. Source and specification control items which include castings and forgings shall be supplied free of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Repair procedure shall not reduce strength or affect function and shall be subject to prior approval by the contracting officer. Impregnated castings shall not be used without specific prior approval of the contracting officer.

3.41.4 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. Reforming of material, before or after welding, shall be subject to prior approval by the contracting officer and shall be supported by test data to show that the method will not compromise the properties of the material or result in built-in stresses.

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Aluminum alloy material or plated surfaces shall be free of water stains, scratches, or scribe lines that exceed 5 percent of the material thickness. Aluminum alloy material shall not be heated for forming or be flame cut. Flame cutting may be employed for steel material providing the original heat treatment characteristics of the material are not affected. All external surfaces shall be free of slivers, burrs, sharp edges and corners, except where specified on the drawings or where they are not detrimental to safety. Fabrication, assembly, and installation shall be accomplished with production tooling, jigs and fixtures to assure that the configuration, alinement, and location of all parts meets the specified requirements. All parts shall be completely and accurately formed to the drawing requirements prior to the bolting, riveting, or welding phases of assembly or installation. Joints, butts and splices shall be limited to those shown on the drawings unless specifically approved by the contracting officer.

3.41.5 Aluminum welds. All aluminum fusion welding shall be in accordance with MIL-W-45205, utilizing the inert gas metal-arc process. The weld types, materials, sizes, and quality of soundness shall be in accordance with the weld requirements specified on the applicable drawings. Class "A" welds shall be stamped to identify the individual welder responsible for the specific welding. Prior to the production fabrication of any MIL-W-45205, Class "A" aluminum weldment, the supplier shall prepare and submit to the contracting officer for approval, surface preparation, cleaning, welding procedures, and workmanship specimens in accordance with MIL-W-45205. When approved by the contracting officer, such procedures and specimens shall be the minimum weld quality acceptable to the Government for Class "A" welds. Prior to the production fabrication of any MIL-W-45205, Class "B" aluminum weldment, the supplier shall prepare surface preparation, cleaning, and welding procedures in accordance with MIL-W-45205 and shall make such procedures available to the Government upon request. All aluminum resistance welding, procedures, and workmanship specimens shall be in accordance with MIL-W-45210.

3.41.6 Steel welds. The weld types, materials, sizes, and quality of soundness shall conform to the weld requirements specified on the applicable drawings. Prior to the production fabrication of any steel weldment, the supplier shall prepare welding procedures and workmanship specimens in accordance with the requirements of MIL-STD-1261, and shall make such procedures available to the Government upon request.

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3.41.7 Welders. Before the assignment of any welder to manual welding of steel or aluminum welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed the prescribed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed in transporter fabrication, and that such qualification is effective as defined by the particular code:

Structural Welding Code, AWS D1.1, Qualification, Section 5.
Welding Qualification of the ASME, Section IX.

The supplier shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes.

3.41.8 Weld repairs. All areas of defective or unacceptable welds shall be repaired in accordance with the weld repair procedures specified in MIL-W-45205, MIL-W-45210, and MIL-STD-1261, as applicable, and shall be subject to prior approval of the contracting officer. Repair of any defective weld shall be limited to two rewelds.

3.41.9 Preformed packing. Where preformed packings are used, no damage to these packings shall occur during installation or assembly. Conditions causing leakage past preformed packings shall be corrected.

3.41.10 Hoses and tubing. All hoses and tubing shall be installed without twists, kinks, or collapsed sections. Tubing bends shall be in accordance with the drawing requirements. Tubing shall not be reformed. Tubing assemblies shall be correctly formed to prevent prestress at installation due to misalinement. Unless otherwise specified herein, hydraulic hose assemblies and their installation shall be in accordance with the requirements of MIL-H-8790. Hydraulic tube flares shall be as specified on the drawings and shall not show evidence of cracks, tool or pit marks, or other defects. Tubes, hoses or their connections shall not leak when tested as specified in 4.6.2.

3.41.11 Bolt torque. Unless otherwise recommended by the component manufacturer, bolts and screws shall be installed in accordance with the torque values specified on the drawings.

3.41.12 Safety wiring. Unless otherwise specified on the drawings, all safety wiring shall be installed in accordance with MS33540.

3.41.13 Sealing compounds. The use of sealing compounds shall be restricted to the types and locations specifically identified on the drawings.

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3.41.14 Interchangeability. All parts, components, and assemblies having the same identification number shall be functionally and dimensionally interchangeable. Interchangeability is defined as two or more parts (components and assemblies) possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts (components and assemblies), except for built-in adjustment, and without selection for fit and performance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, assembled, installed, examined, and tested in accordance with the specified drawings, specifications and standards or as specified in 3.2.3. Incorrect workmanship, materials, dimensions or tolerances shall be cause for rejection.

4.1.1.1 Vendor-supplied components inspection. Vendor-supplied components or assemblies may be inspected and tested at the vendor's facilities, provided said components or assemblies are in full conformance with the specification and drawing requirements.

4.1.1.2 Government-furnished equipment inspection. The supplier's responsibility is limited to the examination of Government-furnished equipment for the verification of interface compatibility with the contract end item. Any incompatibilities, damages, shortages, and malfunctions shall be promptly brought to the attention of the contracting officer.

4.1.2 Disassembly inspection. Failure of any test by the transporter shall be cause for disassembly, in the presence of a Government representative, to the extent necessary to determine the cause of the failure.

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Each disassembled part shall be examined in detail for compliance with this specification and the drawings in regard to workmanship, materials, dimensions, and tolerances. Parts not complying with such requirements shall be rejected. Replacement and reassembly with acceptable material or components, and retesting shall be the responsibility of the supplier.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) First-produced transporter(s) inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of preparation for delivery (see 4.8).

4.3 First-produced transporter(s) inspection.

4.3.1 In-process inspection. During fabrication and assembly of the first-produced transporter(s), the supplier shall conduct a continuing in-process inspection, subject to evaluation and verification by the Government, to assure the supplier's compliance with the dimensions, workmanship, materials, and tolerances specified in the applicable drawings and specifications. Nonconformance to this specification or any applicable drawing shall be cause for rejection of the affected part, component, assembly or transporter, as applicable.

4.3.2 Completed transporter inspection. Upon completion of assembly, one or more first-produced transporters shall be examined for the defects marked "X" in Column 1 of Table III prior to run-in and after run-in and Column 2 of Table III after testing and shall be subjected to the tests marked "X" in Column 1 of Table IV to determine conformance to the requirements of this specification, except as noted in 4.3.2.1. The inspection shall be performed by the supplier and shall be subject to validation or verification by the Government. The test site and facilities shall be subject to approval by the contracting officer.

4.3.2.1 Pretest and posttest examination. Pretest examinations specified herein and in 4.5.1 and Table III shall be performed on individual components/assemblies during fabrication, prior to assembly or during assembly when such examination cannot be performed after completion of assembly. Posttest examinations shall be performed as indicated on those parts requirements applicable to or affected by the test operations.

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4.3.3 Inspection failure. Failure of the first-produced transporter(s) to meet any requirement specified herein due to incorrect workmanship, materials, dimensions, or tolerances; when inspected in accordance with 4.3, shall be cause for rejection or shall be cause for disassembly inspection as specified in 4.1.2, as applicable.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production transporter(s) will be selected at random by the Government from the transporters being produced from production tooling. These transporters will be examined for the defects marked "X" in Column 3 of Table III and subjected to the tests marked "X" in Column 2 of Table IV to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production transporter shall not exclude the remaining transporters from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production transporter to meet any requirements specified herein due to incorrect workmanship, materials, dimensions, or tolerances during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production transporter(s) and shall be cause for refusal by the Government to continue acceptance of production transporters until acceptable evidence has been provided by the supplier that corrective action has been taken to eliminate the defects. Correction of such defects shall be accomplished by the supplier at no cost to the Government on all transporters accepted and produced under the contract. Any defects found as a result of the initial production inspection will be considered prima facie evidence that all transporters accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each transporter shall be examined for the visual defects marked "X" in Column 4 of Table III before the run-in and for functional and visual defects after completion of all testing. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each transporter shall be subjected to the tests marked "X" in Column 3 of Table IV. Failure of any test shall be cause for rejection.

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4.6 Inspection schedule.

4.6.1 Examination. Each transporter shall be examined in accordance with Table III and as specified herein.

Table III. Examination Schedule

First Produced						
Pre-test	Post-test	Init. Prod.	Qual. Conf.	Defect	Inspection Method	Requirement Paragraph
1	2	3	4	5	6	7
X	-	X	X	101. Geometric nonconformance of gaged points.	Government-loaned inspection equipment.	3.1.2, 3.40
X	-	X	X	102. Dimensions and tolerances not as specified.	Measure.	3.1
X	-	X	X	103. Materials not as specified, incorrect heat treatment or protective coatings.	In accordance with referenced material specifications.	3.1, 3.1.1 3.4
X	-	X	X	104. Threads not as specified.	Measure.	3.4
X	X	X	X	105. Lubrication and servicing incomplete or incorrect.	Visual.	3.9, 3.12.2
X	-	X	X	106. Components missing, incorrect, not to form, fit, and functionally interchangeable. Assembly incomplete.	Visual.	3.12.1

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Table III. Examination Schedule (cont'd)

1	2	3	4	5	6	7
X	-	X	X	107. Assembly, installation, or adjustment incorrect.	Visual and functional.	3.12.1
X	X	X	X	108. Instruments function incorrectly or erratically.	Visual and functional.	3.17, 3.29.2, 3.29.3, 3.31.1
X	X	X	X	109. Engine operation, assembly, installation, or adjustment incorrect. Belt tensions or fan shroud adjustment incorrect.	Visual and functional.	3.17
X	X	X	X	110. Engine air induction system installation not as specified. Leakage or damage.	Visual and functional.	3.20
X	X	X	X	111. Cooling system assembly or installation incorrect. Leaks or damaged.	Visual and functional.	3.19
X	X	X	X	112. Exhaust system installation incorrect. Damaged or leaks.	Visual and functional.	3.21
X	X	X	X	113. Fuel system assembly, operation, or installation incorrect. Binding, leakage, damaged, or foreign matter in system.	Visual and functional.	3.18

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Table III. Examination Schedule (cont'd)

1	2	3	4	5	6	7
X	X	X	X	114. Transmission assembly and control assembly installation or adjustment incorrect. Malfunction, leakage or damaged.	Visual and functional.	3.22
X	X	X	X	115. Front axle and engagement control assembly installation or adjustment incorrect. Malfunction, leakage or damaged.	Visual and functional.	3.23
X	X	X	X	116. Parking brake and control assembly installation or adjustment incorrect. Damaged.	Visual and functional.	3.28
X	X	X	X	117. Drive shafts and axle carrier assembly installation or adjustment incorrect. Malfunctions or damaged.	Visual and functional.	3.24
X	X	X	X	118. Wheel and suspension assembly installation or adjustment incorrect. Misalinement, binding, leakage or damage. Incorrect filling or pressurization of shock absorber.	Visual and functional.	3.25

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Table III. Examination Schedule (cont'd)

1	2	3	4	5	6	7
X	X	x	X	119. Land and marine steering system assembly, installation or adjustment not as specified. Misalinement, leakage, binding or damaged.	Visual and functional.	3.26, 3.29.2
X	X	X	X	120. Service brake system installation or adjustment not as specified. Leaks, damaged, or incorrect.	Visual and functional.	3.27
X	X	X	X	121. Electrical system installation not as specified. Slack in cables, malfunction, erratic, limited conductive, discontinuity, damage, inadequate grounding or waterproofing.	Visual and functional.	3.31
X	X	X	X	122. Hydraulic system assembly installation or adjustment not as specified. Leakage, binding, damaged components, cleanliness not as specified, incorrectly tagged, tags missing, overaged hoses, or materials not compatible with specified fluids.	Visual and functional.	3.32

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Table III. Examination Schedule (cont'd)

1	2	3	4	5	6	7
X	X	X	X	123. Pneumatic system assembly, installation or adjustment not as specified. Leakage, binding, damaged, or malfunctions. Belt tension incorrect.	Visual and functional.	3.33
X	X	X	X	124. Marine drive installation assembly, installation, or adjustment not as specified. Leakage, binding, damage, or malfunctions.	Visual and functional.	3.29
X	X	X	X	125. Capstan assembly installation or adjustment not as specified. Leakage or damaged.	Visual and functional.	3.30
X	X	X	X	126. Cab heating system assembly installation or adjustment incorrect.	Visual and functional.	3.35

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Table III. Examination Schedule (cont'd)

1	2	3	4	5	6	7
X	X	X	X	127. Fire extinguisher assembly, installation or adjustment incorrect. Cylinders not fully charged or correctly tagged.	Visual.	3.34
X	X	X	X	128. Treatment and painting not as specified. Incorrect pretreatment, cleaning, or application. Loose, scratched, or incomplete.	Visual.	3.36
X	X	X	X	129. Identification and other markings missing or not as specified.	Visual.	3.37, 3.2.2
X	X	X	X	130. Items to be furnished with transporter missing, or not in correct location.	Visual.	3.38

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Table III. Examination Schedule (cont'd)

1	2	3	4	5	6	7
X	X	X	X	*131. Bolt torque not as specified.	Torque tools.	3.41.11
X	X	X	X	*132. Safety wiring missing or incorrect.	Visual.	3.41.12
X	X	X	X	*133. Sealing compound material or use not as specified.	Visual.	3.41.13
X	X	X	X	*134. Metal fabrication not as specified.	Visual and Measure.	3.41.4
X	X	X	X	135. Welds and weld repairs not as specified, incorrect, missing, incomplete, lack of fusion, burn-through, porosity. (Note: Posttest inspection to be visual examination for soundness of welds.)	In accordance with referenced specifications.	3.41.5, 3.41.6, 3.41.7, 3.41.8
X	X	X	X	136. Grease seals do not restrain lubricant.	Visual.	3.41.2
X	-	X	X	137. Fasteners not as specified.	Visual and measure.	3.1.1
X	X	X	X	138. Misalinement of functional components.	Visual and measure.	3.1.1
X	X	X	X	139. Operating and safety tags missing.	Visual.	3.1.1

* Note: For items 131 thru 134 no disassembly is required.

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Table III. Examination Schedule (cont'd)

1	2	3	4	5	6	7
X	X	X	X	140. Workmanship not as specified, safety hazards.	Visual.	3.41
X	X	X	X	141. Evidence of metallic particles on magnetic drain plugs of engine, transmission, wheel well blower, planetaries, marine drive, capstans and steering housing assemblies.	Visual.	3.9
-	X	X	X	142. Structural damage, deformation or failure.	Visual.	3.7
X	X	X	X	143. Superstructure installation and operation incorrect, incomplete, interference or binding.	Visual and functional.	3.12.3

Note: When **posttest** functional verification is indicated in Column 2, the referenced paragraph in Column 7 need not be repeated. **Posttest** functional verification shall consist only of assurance that all systems, controls, indicators, gages, etc. are still operational, properly adjusted, and visually structurally sound.

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4.6.1.1 Aluminum weld examination. All areas affected by the welding processes shall be examined prior to painting for evidence of defects, including base metal cracks adjacent to or behind welds, arc strikes, cracks in weld metal, undercutting, lack of fusion, porosity, lack of penetration, lack of prescribed fit, incorrect cleaning or preparation for welding, inclusions, incorrect weld size, reduction of adjacent material thickness, or nonadherence to the welding requirements specified in 3.41.5. Weld quality shall be examined and compared against the approved welding specimens submitted. Subassembly welds shall be examined before final installation, wherever feasible. All aluminum welds shall be visually examined with the aid of a 5X magnifying glass. Welds identified on the drawings with the symbol "RT" shall be radiographically examined as specified on the drawings. Evidence of weld defects in excess of that exhibited by the Government-approved welding specimens or the radiographic soundness requirements shall constitute failure of this examination. When radiographic examination indicates that the applicable weld quality requirements have not been met, the Government may require additional radiographic examination of adjacent or other weld areas.

4.6.1.2 Steel weld examination. All areas of welding assemblies affected by the welding processes shall be examined prior to plating, painting, or installation for evidence of defects, including base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting, lack of fusion, porosity, lack of penetration, lack of prescribed fit, incorrect cleaning or preparation for welding, inclusions, incorrect weld size, reduction of adjacent material thickness, or nonadherence to the welding procedures specified in 3.41.6. Weld quality shall be examined and compared against the approved welding specimens submitted. All steel welds shall be visually examined with the aid of a 5X magnifying glass. Steel welds identified on the drawings with the symbol "RT" shall be radiographically examined as specified on the drawings, unless otherwise approved by the contracting officer. Steel welds shall meet the radiographic soundness requirements specified on the applicable drawings. Evidence of weld defects in excess of that exhibited by the approved welding specimens, or the radiographic soundness requirements noted on the applicable drawings shall constitute failure of this examination.

4.6.1.3 Weld repair examination. All weld repair areas shall be visually examined with the aid of a 5X magnifying glass. Weld repairs which evidence defects in excess of that exhibited by the weld specimens, the radiographic soundness requirements required, or the requirements of MIL-W-45205, MIL-W-45210 and MIL-STD-1261, as applicable, shall constitute failure of this examination.

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4.6.2 Tests.

4.6.2.1 Test conditions.

(a) All test apparatus shall be accurate within the limits specified in MIL-STD-448, Section 6, except as specified herein. The transporter instruments may be used to the extent possible for test data readings. All pertinent test data shall be recorded and validated by the supplier and a Government representative. Test data and results shall be incorporated into a test report to be submitted to the Government not later than 60 days after completion of tests for each transporter.

(b) Test methods referenced in tests specified herein are those contained in the appendix.

(c) The transporter shall be serviced in accordance with Drawing 13207E6710 prior to testing. Fuel and lubricants shall be those designated for use in the ambient temperature at the place of test.

(d) Unless otherwise specified in a test, the transporter shall be equipped with an end bay or interior bay superstructure upweighted to the end bay superstructure weight, (approx. 19,500 pounds) plus or minus 100 pounds, and all tests shall be performed at the climatic conditions existing at the place of test.

(e) The transporter shall operate as specified herein without maintenance other than the supplier's recommended normal scheduled maintenance as established by a maintenance schedule proposal prepared and submitted by the supplier to the Government prior to test.

(f) The sequence of testing shall be as follows:

(1) Hydraulic system cleanliness (see 4.6.2.26.2(a)).
(2) Land steering (see 4.6.2.20).
(3) Service brake static (see 4.6.2.21.1).
(4) Service brake dynamic (see 4.6.2.21.2).
(5) Parking brake system (see 4.6.2.22).
(6) Run-in (see 4.6.2.3).
(7) Other supplier-conducted tests in any sequence except 4.6.2.26.2(b) and 4.6.2.33.

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(8) First-produced endurance (see 4.6.2.33).
(9) Hydraulic system cleanliness (see 4.6.2.26.2(b)).

(g) The supplier shall submit a detailed plan of test to the Government for approval, no later than 30 days prior to planned initiation of supplier-conducted tests.

4.6.2.2 Test schedule. Tests shall be in accordance with Table IV.

Table IV. Test Schedule

First Produced	Gov. Tests Init. Prod.	Qual. Conf.	Gov. Tests Insp. Comp.	Test	Primary Test Paragraph	Requirement Paragraph
1	2	3	4	5	6	7
X	-	X	-	Hydraulic system cleanliness.	4.6.2.26.2(a)	3.32.2
X	X	X	X	Land steering.	4.6.2.20	3.26
X	X	X	X	Service brakes (static).	4.6.2.21.1	3.27 3.27.1
X	X	X	X	Service brakes (dynamic).	4.6.2.21.2	3.27.2
X	X	X	X	Parking Brake system.	4.6.2.22	3.28
X	X	X	X	Run-in.	4.6.2.3	3.12.4
X	X	X	X	Speed on level roads.	4.6.2.4	3.13.1
X	X	X	X	Acceleration.	4.6.2.5	3.13.2
X	X	-	X	Drawbar pull.	4.6.2.6	3.13.3
X	X	X	X	Ferry steering.	4.6.2.7	3.14.1
X	X	X	X	Ferry speed.	4.6.2.8	3.14.2
X	X	X	X	Transporter water-tightness.	4.6.2.9	3.15
X	X	X	X	Cab watertightness.	4.6.2.10	3.16
X	X	X	X	Engine system.	4.6.2.11	3.17
X	X	X	X	Fuel system.	4.6.2.12	3.18
X	X	X	X	Cooling system.	4.6.2.13	3.19
X	X	X	X	Engine air intake system.	4.6.2.14	3.20
X	X	X	X	Exhaust system.	4.6.2.15	3.21
X	X	X	X	Transmission system.	4.6.2.16	3.22

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Table IV. Test Schedule (cont'd)

1	2	3	4	5	6	7
X	X	X	X	Front axle engagement.	4.6.2.17	3.23
X	X	X	X	Drive shafts and axle carriers.	4.6.2.18	3.24
X	X	X	X	Wheel and suspension assemblies.	4.6.2.19	3.25
X	-	-	-	Service brakes (fade).	4.6.2.21.3	3.27.2
X	X	X	X	Marine drive system.	4.6.2.23.1	3.29
X	-	-	-	Marine drive static thrust.	4.6.2.23.2	3.29.1
X	X	X	X	Capstans.	4.6.2.24	3.30
X	X	X	X	Battery charging system.	4.6.2.25.1	3.31.1
X	X	X	X	Electrical circuitry.	4.6.2.25.2	3.31.2
X	X	X	X	Headlight alinement.	4.6.2.25.3	3.31.3
X	X	X	X	Windshield wiper system.	4.6.2.25.4	3.31.4
X	X	X	X	Bilge pump system.	4.6.2.25.5	3.31.5
X	X	X	X	Intercommunication system.	4.6.2.25.6	3.31.6
X	X	X	X	Radio communication system.	4.6.2.25.7	3.31.7
X	X	X	X	Hydraulic system performance.	4.6.2.26.1	3.32.1
X	X	X	X	Hydraulic system slaving.	4.6.2.26.3	3.32.3
X	X	X	X	Compressed air system.	4.6.2.27.1	3.33.1
X	X	X	X	Wheel well air blower system.	4.6.2.27.2	3.33.2
X	-	-	-	Fire extinguisher system.	4.6.2.28	3.34
X	X	X	X	Cab heating system.	4.6.2.29	3.35
X	X	X	X	Treatment and painting.	4.6.2.30	3.36
X	-	-	-	Electromagnetic interference.	4.6.2.31	3.8

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Table IV. Test Schedule (cont'd)

1	2	3	4	5	6	7
-	-	-	-	Environmental. 1/	4.6.2.32	3.5
X	-	-	-	Endurance.	4.6.2.33	Section 3
-	X	-	X	Operational.	4.6.2.34	Section 3
X	X	X	X	Hydraulic system cleanliness.	4.6.2.26.2(b)	3.32.2
X	X	X	X	Lifting eye.	4.6.2.35	3.7
X	X	X	X	Towing eye.	4.6.2.36	3.7

1/ When specified.

4.6.2.3 Run-in. Prior to any other testing, each transporter shall be run-in in accordance with Test Method No. 1 (see Appendix). Inability of the transporter to complete the break-in run shall constitute failure of this test.

4.6.2.4 Speed on level roads. Operate the transporter on smooth, level, hard-surfaced roads at maximum safe speed for a period of at least 15 minutes. Speed and distance shall be measured using the cab panel mounted speedometer. Nonconformance to 3.13.1 shall constitute failure of this test.

4.6.2.5 Acceleration. With the transporter on a smooth, level, hard-surfaced road, accelerate from 0 to 35 mph. Repeat for a total of three runs in each direction over the same course. Record the time to attain a speed of 35 mph for each run. Nonconformance of the average of the six trials to 3.13.2 shall constitute failure of this test.

4.6.2.6 Drawbar pull. Determine the drawbar pull of the transporter at stall, 0.5 mph, and 15 mph when operating in the 1-2 or 3-4 transmission gear range and in four-wheel drive. Make three runs in each direction over the same course. Nonconformance of the average value of these runs to 3.13.3 shall constitute failure of this test.

4.6.2.7 Ferry steering. Assemble a two-unit ferry. While operating both units at full throttle, steer the ferry from the cab of one unit 180 degrees hard to left, then 180 degrees hard to right. Repeat this maneuver twice, from the cab and rear control station of each unit, one at a time. Next, repeat the above cycle while steering 360 instead of 180 degrees. Nonconformance to 3.14.1 shall constitute failure of this test.

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